LESSON 6.10

Divide Mixed Numbers

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can explain what dividing with mixed numbers means, rewrite them as improper fractions, find the quotient with common denominators or by multiplying by the reciprocal, and check it by multiplying.

WHY THIS MATTERS Agent Chen is mapping a suspect's route. The total distance is 3 1/2 miles, and each segment between landmarks is 1/4 mile. She needs to know how many segments are on the route to place surveillance cameras at every landmark.

REMEMBER

Divide Mixed Numbers

Dividing Mixed Numbers A mixed number, like 3 1/2, has a whole part and a fraction part. Before you can divide, you change each mixed number into an improper fraction (top heavier than bottom). Then ask how many of the divisor fit — with common denominators or by multiplying by the reciprocal — and check by multiplying the quotient by the divisor.

1. Convert each mixed number to an improper fraction: 3 1/2 = 7/2.
2. Name the question and divide: 7/2 ÷ 1/4 asks how many 1/4s fit in 7/2. With common denominators, 7/2 = 14/4, and 14 fourths ÷ 1 fourth = 14, or multiply by the reciprocal: 7/2 × 4/1 = 28/2 =
3. Simplify, then check by multiplying: 14 × 1/4 = 14/4 = 3 1/2

Words: **Mixed Number *(Número mixto)*** **Improper Fraction *(Fracción impropia)*** **Convert *(Convertir)*** **Simplify *(Simplificar)*** **Reciprocal *(Recíproco)***

PART 1 Practice the skill

show every step

1 **Circle the letter of the best answer. Show how you know.**

What is 2/3 ÷ 1/3?

(A) 1/2 (B) 2/9

(C) 2 (D) 6

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

What is 1/2 ÷ 1/4?

(A) 1/8
(B) 1/2
(C) 2
(D) 4

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

What is 3/4 ÷ 1/8?

(A) 3/32
(B) 6
(C) 8/3
(D) 24

SHOW YOUR WORK

PART 2 Apply it

real-world problems

4 **Circle the letter of the best answer. Show how you know.**

A 4 1/2-cup jar of fingerprint gel fills exactly 1 1/2 test kits. How much gel is in ONE kit? (4 1/2 ÷ 1 1/2 asks how much is in one whole kit.)

(A) 6 3/4 cups
(B) 1/3 cup
(C) 3 cups
(D) 4 1/3 cups

HINT The jar fills MORE than one kit, so one kit holds less than 4 1/2 cups.

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

What is the reciprocal of 3/5?

(A) 1/3
(B) 3/5
(C) 5/5
(D) 5/3

Rewrite each fraction _____

Work it out _____

Simplify _____

1. **Same size pieces** *Common denominator first.*
2. **Operate** *Only then add or subtract.*
3. **Simplify** *Divide out common factors.*

6 **Circle the letter of the best answer. Show how you know.**

A detective has a rope that is 5 1/4 feet long. She needs to cut it into pieces that are each 3/4 of a foot for tying evidence tags. How many pieces can she cut? Write 5 1/4 as an improper fraction.

(A) 9/4
(B) 20/4
(C) 21/4
(D) 51/4

Write any whole number over 1 _____

Keep · Change · Flip _____

Multiply across _____

Simplify · label the units _____

1. **Whole over 1** *3 becomes 3/1.*
2. **Keep** *The first fraction does not move.*
3. **Change · Flip** *÷ becomes ×; flip the second fraction.*
4. **Check** *Dividing by a piece smaller than 1 gives a BIGGER answer.*

EXPLAIN YOUR THINKING

Why is converting to an improper fraction the first step when dividing mixed numbers?

SENTENCE STARTER Before dividing, I convert 2 1/2 to ___ because it is easier to ___ with improper fractions.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

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LEARNING TARGET I can explain what dividing with mixed numbers means, rewrite them as improper fractions, find the quotient with common denominators or by multiplying by the reciprocal, and check it by multiplying.

WHY THIS MATTERS Agent Chen is mapping a suspect's route. The total distance is 3 1/2 miles, and each segment between landmarks is 1/4 mile. She needs to know how many segments are on the route to place surveillance cameras at every landmark.

PART 1 Understand the idea

models and meaning

1 **Write the letter of the matching item on each line.**

Mission 4: Match each suspect's case-file number (a mixed number) to its decoded improper fraction.

_____ 5 1/4	(A) 21/4
_____ 2 3/5	(B) 13/5
_____ 3 1/6	(C) 19/6
_____ 4 2/3	(D) 14/3

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

A detective has a rope that is 5 1/4 feet long. She needs to cut it into pieces that are each 3/4 of a foot for tying evidence tags. How many pieces can she cut? 5 1/4 ÷ 3/4 = ?

(A) 3 3/4 (B) 4
(C) 6 (D) 7

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

A detective has a rope that is 5 1/4 feet long. She needs to cut it into pieces that are each 3/4 of a foot for tying evidence tags. How many pieces can she cut? Check the answer: how long are 7 pieces of 3/4 foot?

(A) 3/4 foot
(B) 5 1/4 feet
(C) 5 3/4 feet
(D) 7 feet

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Mission 1: The stakeout shift is 4 1/2 hours long. Each agent covers a 1 1/2-hour watch. How many agent watches fill one shift?

(A) 2 1/2
(B) 3
(C) 4 1/2
(D) 6

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

Mission 2: A surveillance route is 3 miles long, split into 1 1/2-mile patrol sections. How many sections are on the route?

(A) 1 1/2
(B) 2
(C) 3
(D) 4 1/2

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

Mission 3: There are 7 1/2 feet of caution tape to seal a crime scene. Each doorway needs 2 1/2 feet. How many doorways can be sealed?

(A) 2 1/2
(B) 3
(C) 5
(D) 10

SHOW YOUR WORK

EXPLAIN YOUR THINKING

Why is converting to an improper fraction the first step when dividing mixed numbers?

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